Jiayi Sun's Astro Colloquia Notes | October 27, 2026 | Zahi Rahman

Undergrad at Tsinghua University in China, visiting Postdoc at UIUC and accord to Prof. Wong has a knack for turning unique, original project ideas to those with actual results.

Studying the Interstellar Medium (ISM) = gas and dust between stars and star clusters. Star and Star clusters form from a "large ensemble" from the molecular ISM. These inject mass, momentum, and energy back into the ISM. This can actually reshape the structure of the ISM as shown in the image. This formes a cycle from the gas to the stellar formation.

Core Question: "Given the current state of the gas, can we explain and predict the star formation rate" - Not easy to accurately constrain, so researchers have broken it into smaller problems.

Phangs: collaboration "Physics at High Angular resolution in Nearby Galaxies", started from ALMA initiative. Around 100+ members and 170 peer-reviewed publications (~at least 1 paper per member which is "pretty good"). Extended effort with HST and JWST data for optical, and mid and near IR imaging, on top of many other "synergistic" datasets. Each dataset really gives information across different bands, like how Chandra, a latest addition, gives more information from SNR and X-ray binaries.

$$SFR \propto M_{gas}$$

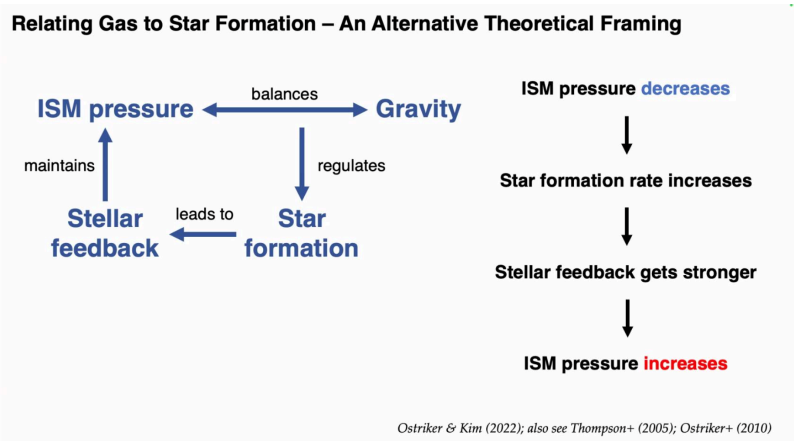
More than 90% of the total molecular gas is H2, and due to conditions in the ISM this is typically rare to observe H2 emission. Halpha and other lines from HII regions are powered by young massive stars.You can use UV, mid-IR, and radio observation to trace that star formation rate. Molecular Kaneshkar relation shows SFR trace (surface density) vs. Gas with a linear slope. The residual scatter is a factor of around 2, 0.3 dex, and the depletion time is 1 to 3 gigayears which are upper limits. The plot shows each data point as each one kpc chunk of the galaxy. They choose kpc scale because the tracers are from samples from the galaxy so they need samples from many different molecular cloud regions of varying masses.

$$SFR = \varepsilon_{ff} \frac{M_{gas}}{t_{ff}}$$

They use the theoretical formula for free fall star formation where the time for formation due to gravitational collapse rather than any other method, and then empirically it is multiplied by that efficiency with the median value being around only 1%. This is interesting because it was talked about on Monday, for Astro 406 for the same exact sorts of contexts around stellar formation. This efficiency is so low is believed to be due

to the “gas dynamical state”, and “stellar feedback”.This is when pressure in the gas (turbulence and the magnetic field) counteracts the effects of gravity, and how new stars irradiate and “shock” surrounding gas to stop star formation. The majority of the talk centered here, where he answered technical questions on the validity of using this free fall timescale given that there have been no ordered velocities correlating to this freefall observed in our galaxy, to which he answered that it is a model for star formation that serves as a benchmark for improvement, and other models of this star formation would need to be tested by simulation researchers to find “better” timescales to constrain.There is an observed slight positive correlation between the efficiency of free fall, epsilon ff, and alpha.

There is an alternative framing from gas to stellar formation. This is much more physically involved, better understanding the ISM pressure from the dynamical state of gas, and turbulence from stellar feedback (mostly from supernovae as it produces the strongest effects). This is affected by stellar formation itself, so the ISM pressure, gravity, star formation, and stellar feedback form a closed loop that affects one another. An image of this is shown below:



This is coined a “Pressure-regulated Feedback-modulated” Star Formation relation. The observed relation of ISM equilibrium pressure and SFR surface density agree well with the predictions from this theoretical relation. The ratio of these terms is the “feedback yield” which is the amount of energy and momentum injected into the ISM by feedback.

Summary:

- PHANGS is a multiwavelength survey that probes over 100k SFR across all evolutionary stages in the universe, finding:
    - SF is inefficient on cloud scales -> 1% efficiency for free fall time
    - The efficiency above is correlated with the gas dynamical state, turbulence from pressure of gas and supernovae, which contradicts the cloud-scale SF models

- ISM pressure correlates with SFR on kpc scales, as predicted by the Pressure-regulated Feedback-modulated theory, and the ratio of ISM pressure and SFR surface density is consistent with supernovae turbulence effects.

Thank you, I hope this qualifies for Extra Credit for ASTRO 457.